- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. Unless otherwise instructed, when a numerical value is needed, use $g = 9.8$.

9 A pebble is thrown horizontally at $14\,\text{m s}^{-1}$ from a window which is 5 m above horizontal ground. The pebble goes over a fence 2 m high d m away from the window as shown in Fig. 9. The origin is on the ground directly below the window with the x-axis horizontal in the direction in which the pebble is thrown and the y-axis vertically upwards.

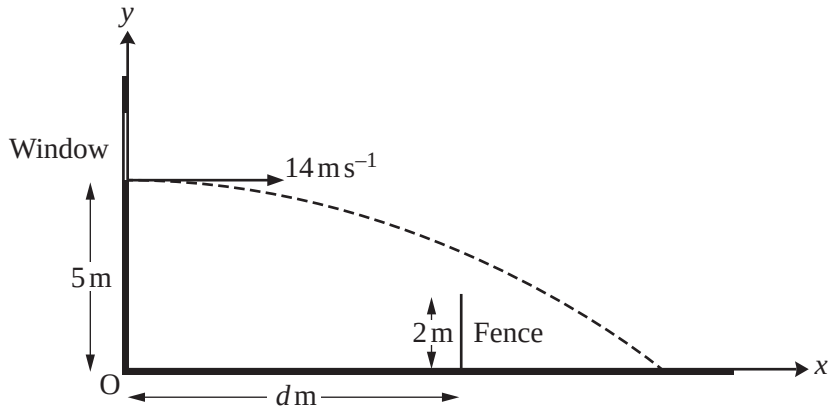


Fig. 9

(i) Find the time the pebble takes to reach the ground. **[3]**

(ii) Find the cartesian equation of the trajectory of the pebble. **[4]**

(iii) Find the range of possible values for d. **[3]**